Li Jing

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Research Interests

Machine Learning
Self-supervised Learning
Computer Vision
AI for Science

Education

Massachusetts Institute of Technology — Cambridge, MA
Ph.D. in Physics — 2014 - 2020
(interdisciplinary area of Machine Learning and Physics)
Dissertation: Physical Symmetry Enhanced Neural Networks

Peking University — Beijing, China
B.S. in Physics — 2010 - 2014

Research Experience

Facebook AI Research — New York, NY
Postdoctoral Researcher — 2020 - Present
Advisor: Yann LeCun

- Fundamental Research on Self-supervised Learning
  - proposed BarlowTwins, a state-of-the-art self-supervised learning framework for visual representation learning. This work remains one of the highest impact research in 2021, which has received 350+ citations and 750+ Github stars.
  - theoretically studied the dynamics in self-supervised learning, featured in Meta AI Blog and NeurIPS self-supervised learning tutorial.
  - proposed several techniques for applying self-supervised learning, including (a) equivariant SSL, on adapting domain knowledge; (b) positive-unlabeled SSL, on utilizing biased label information; (c) progressive fine-tuning, on generalizing pretrained models to out-of-distribution datasets.
- Self-supervised Learning Applications
  - applied SSL methods to EMG data, which reformed the data collection pipeline and advanced production performances (collaboration with Meta Reality Labs CTRL Team)
  - applied SSL methods to improve mobile vision model efficiency, robustness, and privacy. (collaboration with Meta Reality Labs Mobile Vision Team)

Massachusetts Institute of Technology — Cambridge, MA
Research Assistant — 2016 - 2020
Advisor: Marin Soljacic

- Physical Symmetry Enhanced Neural Networks

- proposed an efficient unitary recurrent neural network framework based on Givens rotation, which achieved state-of-the-art performances on long-term memory time-series benchmarks. This work has had a broad impact on the optics and quantum computing community.
- proposed gated orthogonal recurrent neural networks, which demonstrated state-of-the-art performances on NLP applications requiring long-term memory.

- AI for Photonics
  - first proposed to use neural networks to replace FDTD simulation, exponentially speed up forward simulation and backup design.
  - proposed transfer learning techniques for photonics simulation, allows data expensive scenarios to leverage easy access data.

PUBLICATIONS

Google Scholar: `https://scholar.google.com/citations?user=VhxDLwcAAAAJ`

Peer Reviewed Machine Learning Conference Papers

1. **Li Jing**, Pascal Vincent, Yann LeCun, Yuandong Tian, "Understanding Dimensional Collapse in Contrastive Self-supervised Learning", **International Conference on Learning Representations (ICLR)** (2022)
2. Rumen Dangovski, **Li Jing**, Charlotte Loh, Seungwook Han, Akash Srivastava, Brian Cheung, Pulkit Agrawal, Marin Soljacic , "Equivariant Contrastive Learning", **International Conference on Learning Representations (ICLR)** (2022)
3. Jure Zbontar*, **Li Jing***, Ishan Misra, Yann LeCun, Stéphane Deny, "Barlow Twins: Self-Supervised Learning via Redundancy Reduction", International **Conference on Machine Learning (ICML)** (2021)
4. Rumen Dangovski, Michelle Shen, Dawson Byrd, **Li Jing**, Desislava Tsvetkova, Preslav Nakov, Marin Soljacic, "We Can Explain Your Research in Layman's Terms: Towards Automating Science Journalism at Scale", **AAAI Conference on Artificial Intelligence (AAAI)** (2021)
5. **Li Jing**, Jure Zbontar, Yann LeCun, "Implicit Rank-Minimizing Autoencoder", **Conference on Neural Information Processing Systems (NeurIPS)** (2020)
6. Matthew Khoury, Rumen Dangovski, Longwu Ou, Yichen Shen, Preslav Nakov, **Li Jing**, "Vector-Vector-Matrix Architecture: A Novel Hardware-Aware Framework for Low-Latency Inference in NLP Applications", **Conference on Empirical Methods in Natural Language Processing (EMNLP)** (2020)
7. **Li Jing***, Yichen Shen*, Tena Dubček, John Peurifoy, Scott Skirlo, Yann LeCun, Max Tegmark, Marin Soljacic, "Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs", **Proceedings of International Conference on Machine Learning (ICML)** (2017)

Machine Learning Journal Papers

8. Zhedong Wang, Chao Qian, Tong Cai, Longwei Tian, Zhixiang Fan, Jian Liu, Yichen Shen, **Li Jing**, Jianming Jin, Er-Ping Li, Bin Zheng, Hongsheng Chen, "Demonstration of Spider-Eyes-Like Intelligent Antennas for Dynamically Perceiving Incoming Waves", **Advanced Intelligent Systems** (2021)

9. Samuel Kim, Peter Lu, Srijon Mukherjee, Michael Gilbert, **Li Jing**, Vladimir Ceperic, Marin Soljacic, "Integration of Neural Network-Based Symbolic Regression in Deep Learning for Scientific Discovery" **IEEE Transactions on Neural Networks and Learning Systems (TNNLS)** (2020)

10. Rumen Dangovski*, **Li Jing***, Marin Soljacic, "Rotational Unit of Memory: A Novel Representation Unit for RNNs with Scalable Applications", **Transactions of the Association for Computational Linguistics (TACL)** (2019)

11. **Li Jing***, Caglar Gulcehre*, John Peurifoy, Yichen Shen, Max Tegmark, Marin Soljacic, Yoshua Bengio, "Gated Orthogonal Recurrent Units: On Learning to Forget", **Neural Computation** (2019)

Physics Journal Papers

12. Thomas Christensen, Charlotte Loh, Stjepan Picek, Domagoj Jakobovic, **Li Jing**, Sophie Fisher, Vladimir Ceperic, John Joannopoulos, Marin Soljacic, "Predictive and generative machine learning models for photonic crystals", **Nanophotonics** (2020)

13. Mihika Prabhu, Charles Roques-Carmes, Yichen Shen, Nicholas Harris, **Li Jing**, Jacques Carolan, Ryan Hamerly, Tom Baehr-Jones, Michael Hochberg, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljacic, "A Recurrent Ising Machine in a Photonic Integrated Circuit", **Optica**, (2020)

14. Chao Qian, Bin Zheng, Yichen Shen, **Li Jing**, Erping Li, Lian Shen, Hongsheng Chen, "Deep learning enabled self-adaptive invisibility cloak", **Nature Photonics** (2020)

15. Charles Roques-Carmes, Yichen Shen, Cristian Zanoci, Mihika Prabhu, Fadi Atieh, **Li Jing**, Tena Dubcek, Vladimir Ceperic, John Joannopoulos, Dirk Englund, Marin Soljacic, "Heuristic Recurrent Algorithms for Photonic Ising Machines", **Nature Communications** (2020)

16. Yurui Qu*, **Li Jing***, Yichen Shen, Min Qiu, Marin Soljacic, "Migrating Knowledge between Physical Scenarios based on Artificial Neural Networks", **ACS Photonics** (2019)

17. John Peurifoy, Yichen Shen, **Li Jing**, Yi Yang, Fidel Cano-Renteria, Brendan Delacy, John Joannopoulos, Max Tegmark, Marin Soljacic, "Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks", **Science Advances** (2018)

18. Heng Fan, Yi-Nan Wang, **Li Jing**, Jie-Dong Yue, Han-Duo Shi, Yong-Liang Zhang, Liang-Zhu Mu, "Quantum cloning machines and the applications", **Physics Reports** (2014)

19. Yong-Liang Zhang, Huan Wang, **Li Jing**, Liang-Zhu Mu, Heng Fan, "Fitting magnetic field gradient with Heisenberg-scaling accuracy", **Scientific Reports** (2014)

20. Yong-Liang Zhang, Yi-Nan Wang, Xiang-Ru Xiao, **Li Jing**, Liang-Zhu Mu, VE Korepin, Heng Fan, "Quantum network teleportation for quantum information distribution and concentration", **Physical Review A** (2013)

21. **Li Jing**, Yi-Nan Wang, Han-Duo Shi, Liang-Zhu Mu, Heng Fan, "Minimal input sets determining phase-covariant and universal quantum cloning", **Physical Review A** (2012)

22. Zhao-Xi Xiong, Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Jin Lei, Liang-Zhu Mu, Heng Fan, "General quantum key distribution in higher dimension", **Physical Review A** (2012)
23. Yi-Nan Wang, Han-Duo Shi, **Li Jing**, Zhao-Xi Xiong, Jin Lei, Liang-Zhu Mu, Heng Fan, "Non-compression of quantum phase information", **Journal of Physics A: Mathematical and Theoretical** (2012)
24. Yi-Nan Wang, Han-Duo Shi, Zhao-Xi Xiong, **Li Jing**, Xi-Jun Ren, Liang-Zhu Mu, Heng Fan, "Unified universal quantum cloning machine and fidelities", **Physical Review A** (2011)

Pre-prints

25. **Li Jing***, Jiachen Zhu*, Yann LeCun, "Masked Siamese ConvNets", under review of NeurIPS 2022
26. Anish Acharya, **Li Jing**, Sujay Sanghavi, Bhargav Bhushanam, Dhruv Choudhary, Michael Rabbat, Inderjit Dhillon, "Positive Unlabeled Contrastive Learning", under review of NeurIPS 2022
27. Andrew Ma, Yang Zhang, Thomas Christensen, Hoi Chun Po, **Li Jing**, Liang Fu, Marin Soljacic, "Topogivity: A Machine-Learned Chemical Rule for Discovering Topological Materials ", arXiv: 2202.05255, under review of Science Advances
28. Ileana Rugina, Rumen Dangovski, Renbin Liu, **Li Jing**, Preslav Nakov, Marin Soljacic, "Data-Informed Global Sparseness in Attention Mechanisms for Deep Neural Networks", arXiv: 2012.02030
29. Evan Vogelbaum, Rumen Dangovski, **Li Jing**, Marin Soljačić, "Contextualizing Enhances Gradient Based Meta Learning", arXiv: 2007.10143
30. **Li Jing***, Rumen Dangovski*, Marin Soljacic, "WaveletNet: Logarithmic Scale Efficient Convolutional Neural Networks for Edge Devices", arXiv:1811.11644
31. Han-Duo Shi, Yi-Nan Wang, **Li Jing**, Ru-Quan Wang, Liang-Zhu Mu, Heng Fan, "Quantum key distribution based on a quantum retrodiction protocol", arXiv: 1205.3556

Patents

32. Charles Roques-armes, Yichen Shen, **Li Jing**, Tena Dubcek, Scott A Skirlo, Hengameh Bagherianlemraski, Marin Soljacic, *"Optical Ising machines and optical convolutional neural networks"*
US patent number 11,017,309, issued in May 2021

INVITED TALKS

1. Towards Scalable Self-supervised Learning
Salesforce Research Virtual, 2022
2. Understanding and Solving Dimensional Collapse in Contrastive SSL
Flatiron Research Institute New York, NY, 2022
Center for Computational Mathematics
3. Understanding and Solving Dimensional Collapse in Contrastive SSL
UC Davis Virtual, 2022
AI, Robotics and Automation Seminar
4. Implicit Regularization and Self-supervised Learning
OpenAI Virtual, 2021

5. On Dimensional Collapse in Contrastive Self-supervised Learning
Google Research Virtual, 2021
6. Barlow Twins: Self-supervised Learning via Redundancy Reduction
International Conference on Machine Learning (ICML) Virtual, 2021
7. Implicit Rank-Minimizing Autoencoder
Conference on Neural Information Processing Systems (NeurIPS)
Virtual, 2020
8. Physics Inspired Deep Learning Algorithms
Facebook AI Research New York, NY, 2019
9. Phase-encoded Recurrent Neural Networks and their Applications
DeepMind Mountain View, CA, 2019
10. Phase-encoded RNNs and Scientific Text Summarization
Amazon Alexa AI Cambridge, MA, 2019
11. Photonics and AI: Algorithms and Applications
APS Physics of AI conference Yorktown Heights, NY, 2019
12. **MIT-CHIEF Annual Conference, AI panel** Cambridge, MA, 2018
13. Nanophotonic Particle Simulation and Inverse Design Using Artificial Neural Networks
Conference on Neural Information Processing Systems (NIPS)
Deep Learning for Physical Science Workshop Long Beach, CA, 2018
14. Gated Orthogonal Recurrent Units: On Learning to Forget
AAAI Conference on Artificial Intelligence (AAAI)
Workshop on Reasoning and Learning for Human-Machine Dialogues
New Orleans, LA, 2018
15. Tunable Efficient Unitary Neural Networks (EUNN) and their application to RNNs
International Conference on Machine Learning (ICML)
Sydney, Australia, 2017

PRESS COVERAGE

1. Self-Adaptive Invisibility Cloak: The "Magic" in Optics.
Techacute 2020
2. Solving complex problems at the speed of light.
SciTechDaily 2020
3. Lightelligence releases prototype of its optical AI accelerator chip
Venturebeat 2019
4. Can science writing be automated? A neural network can read scientific papers and render a plain-English summary.
MIT News 2019
5. AI-based method could speed development of specialized nanoparticles. Neural network could expedite complex physics simulations.
MIT News 2018

ACADEMIC SERVICES

Conference reviewer

- International Conference on Learning Representations (ICLR)

- International Conference on Machine Learning (ICML)
- Conference on Neural Information Processing Systems (NeurIPS)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)

Journal reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Computational Intelligence Magazine (CIM)
- IEEE Photonic Technology Letters
- Proceedings of the National Academy of Sciences (PNAS)
- Physical Review Letters
- Physical Review A
- Physical Review B
- Physical Review X
- Physical Review Research
- Physical Review Applied
- Scientific Reports
- Journal of Applied Physics
- Optics Express
- Optica
- Advanced Photonics
- Nanophotonics
- ACS Photonics
- Nature Photonics

Mentoring

1. Anish Acharya — 2021
   Ph.D at UT Austin, intern at Facebook
2. Jiachen Zhu — 2021
   Ph.D at New York University, intern at Facebook
3. Pawan Goyal — 2019
   Undergrad at MIT
4. Lay Jain — 2019
   Undergrad at MIT, now quantitative researcher at Citadel Securities
5. Alvaro Inesta — 2018
   Exchange Master student at MIT
6. Ivan Ivanov — 2017
   RSI program at MIT, now undergrad at Cambridge University
7. Rawn Henry — 2017
   Undergrad at MIT, now AI Engineer at Nvidia
8. John Peurifoy — 2017
   Undergrad at MIT, now CEO at Floating point group
9. Rumen Dangovski — 2017
   Undergrad at MIT, now Ph.D student at MIT EECS

SELECTED HONORS AND AWARDS

- Forbes China 30 under 30 - Enterprise Technology 2019
- Peking University Outstanding Graduate 2014
- Peking University Academic Innovation Award 2013
- International Physics Olympiad (IPhO) - Gold Medal 2010

INDUSTRIAL EXPERIENCE

Lightelligence Inc. Boston, MA
Co-founder, Chief Algorithm Architect 2018 - 2019
MIT Spin-out, Optical AI Accelerator Startup